Coset intersections under group operations

Human Compression Project

Abstract

Common representatives of incident cosets provide a uniform language for free products, coherent amalgams, HNN extensions and orbit constructions. We prove separate transfer theorems for tree-supported cosets and semidirect-product fibers, retaining nontrivial chamber stabilizers. Exact panel and residue formulas distinguish the hypotheses needed for firmness, thinness and residual connectedness. Cyclic markings and inverse presentation maps give the Artin applications, while orbital presentations and module changes of coordinates retain the complete shared theory used by the diagonal and wreath constructions. The accompanying application applies these results to Piedade and Tranchida's 2025 paper with explicit source qualifications.

The common thread is the existence of a representative for every finite family of pairwise intersecting parabolic cosets. On a tree, representatives are transported through full subcomplexes or compatible local cosets; in a semidirect product they are assembled in a common second-coordinate fiber. These mechanisms have different hypotheses. No theorem about arbitrary graphs of geometries follows simply by combining them.

This is a complete written theory paper for three existing application contracts: N005 (amalgamation, HNN extensions and twisting), N006 (wreath and semidirect constructions) and N007 (element-diagonal constructions). Its modular sections have explicit uses. The signed-module section is used by N006, not by N005; the tree and cyclic sections supply the additional N005 routes. Full relevant common theory remains part of any eventual comparison. This draft contains no length comparison or new theorem-credit claim.

1 Cosets, flags and residue generation

All type sets in this paper are finite. Cosets are left cosets. In a coset geometry $(B, (B_i)_{i\in I})$ write $B_J = \bigcap_{i\in J} B_i$, with $B_\emptyset = B$. A regular hypertope is thin, residually connected and flag-transitive. Automorphisms preserve types. In its faithful standard model, the group acts freely and transitively on chambers. Its distinguished involutions b_i interchange a fixed chamber and its i-adjacent chamber; then

$$B_J = \langle b_i : i \in I \setminus J\rangle, \qquad B_I = 1. \tag{1}$$

We recall why the geometric hypotheses give these assertions. In a thin geometry a type-preserving automorphism fixing a chamber fixes all its adjacent chambers. Residual connectivity implies that the chamber graph, and the chamber graph of every residue, is connected: by induction on rank, chambers containing a given element are connected; a path in the incidence graph connects these families, using chambers containing each consecutive incident pair. The rank-one and rank-two cases start the induction. Thus a chamber stabilizer is trivial. The same argument inside the residue of the base flag of types J proves (1). It also proves that the full type-preserving automorphism group is the displayed group whenever a faithful group already acts transitively on chambers.

For a second regular hypertope $(A, (A_x)_{x\in S})$, put $A(T) = \langle a_x : x \in T\rangle$ for $T \subseteq S$. Formula (1) gives

$$A(T) \cap A(T') = A(T \cap T'). \tag{2}$$

We will use both this intersection property and flag-transitivity. They are different requirements: the intersection property alone is not a general criterion for a nonstring coset system to be a geometry.

Lemma 1. *A coset incidence system $(A, (A_i)_{i\in I})$ is flag-transitive if and only if the family of all left cosets of all A_J, $J \subseteq I$, is Helly.*

Proof. A flag is a collection of pairwise intersecting cosets of distinct maximal parabolics. It is a translate of a standard flag precisely when these cosets have a common element. This proves one implication. For the other, expand each coset aA_J into the intersection of aA_j, $j \in J$. If two original cosets intersect, every maximal coset from one meets every maximal coset from the other. Within one expanded coset they meet at a. Two such cosets of the same maximal parabolic are therefore equal. The resulting distinct-type collection is a flag, and flag transitivity supplies a common element. Empty J imposes no constraint. A standard flag extends to the standard chamber, so flag transitivity also supplies the geometric property. □

Lemma 2 (Coset intersection)**.** *If B is flag-transitive on its coset geometry, every finite pairwise-intersecting family of cosets of standard parabolic subgroups has nonempty total intersection. In particular, for $H, K, J \subseteq I$,*

$$B_H B_J \cap B_K B_J = B_{H\cup K} B_J.$$

Proof. Lemma 1 gives the first assertion. For the last assertion, if $g \in B_H B_J \cap B_K B_J$, the three cosets B_H, B_K, gB_J intersect pairwise. A point of their common intersection belongs to $B_{H\cup K}$ and to gB_J. The reverse inclusion follows from subgroup containment. □

Theorem 3 (Element orbits)**.** *Fix $j \in I$ and $x_0 = B_j \in X = B/B_j$. If B is flag-transitive, then*

$$(B_H x_0) \cap (B_K x_0) = B_{H\cup K} x_0 \qquad (H, K \subseteq I). \tag{3}$$

Thus the complementary notation $O(M) = B_{I\setminus M} x_0$ satisfies $O(M) \cap O(N) = O(M \cap N)$.

Proof. The inverse image of $B_H x_0$ under $B \to B/B_j$ is $B_H B_j$. Apply Lemma 2 and take images. Finally $(I \setminus M) \cup (I \setminus N) = I \setminus (M \cap N)$. □

1.1 Parabolics with a nontrivial chamber stabilizer

We now allow flag-transitive coset geometries whose acting groups are not faithful. The type sets remain finite and nonempty. In a geometry $\alpha = (A, (A_x)_{x\in S})$ put $A_E = \bigcap_{x\in E} A_x$, with $A_\varnothing = A$, and call A_S the Borel subgroup. It need not be trivial. The residue of the standard flag of types E is the coset geometry of A_E with parabolics $A_{E\cup\{x\}}$, $x \notin E$. To see this identification, take an element incident with that flag; flag transitivity supplies a common representative in A_E. Two such representatives define the same element exactly when their quotient lies in $A_E \cap A_x$. This also identifies incidence by intersecting the corresponding cosets.

For a coset geometry of rank at least two the incidence graph is connected if and only if its maximal parabolics generate the acting group. Indeed, if H is that generated subgroup, paths from the standard flag remain among cosets with representatives in H. Conversely multiplication successively by elements of the parabolics provides paths through incident cosets; when two successive steps have the same type, an intervening different type gives the same conclusion. Thus the components are the translates of this component. Applied to every residue, this proves that residual connectedness is equivalent to

$$A_E = \langle A_{E\cup\{x\}} : x \notin E\rangle \quad \text{if } |S \setminus E| \geq 2. \tag{4}$$

For rank one residual connectedness imposes no condition.

Lemma 4 (Generation of parabolics)**.** *If α is residually connected, then*

$$A_E = \langle A_S,\ A_{S\setminus\{x\}} : x \notin E\rangle.$$

Consequently, for any nonempty family (E_ν) of subsets of S,

$$\langle A_{E_\nu} : \nu\rangle = A_{\bigcap_\nu E_\nu}. \tag{5}$$

Proof. Induct on $|S\backslash E|$. For zero this is $A_S = A_S$; for one the displayed generator is A_E itself. For larger complements, use (4) and the induction hypothesis on each child. Every missing type remains missing in at least one child, so all and only the indicated minimal parabolics occur. For the second assertion, each group on the left lies in the right-hand group. Conversely, every minimal parabolic required to generate the right-hand group occurs in some A_{E_v}, and A_S lies in every one. The nonempty-family hypothesis is necessary when $A_S \neq 1$. □

The explicit A_S in the first formula must not be dropped at $E = S$. This avoids the empty-generated-subgroup problem in the source's background equivalences.

2 Tree gluing and coherent amalgamation

An abstract simplicial complex is *flag* if each finite set of pairwise adjacent vertices spans a simplex. A subcomplex is *full* if it contains every simplex of the ambient complex whose vertices it contains. We first isolate a standard elementary gluing principle and prove the form needed here. No novelty is claimed for the principle itself.

Lemma 5. *Let T be a tree. For each vertex v let X_v be a flag simplicial complex. For each edge $e = uv$ let X_e be a complex equipped with injective simplicial maps to X_u and X_v, both with full image. Form X by identifying these images. Then each X_v embeds as a full subcomplex of X, and X is flag.*

Proof. An identification of vertices is a finite sequence of identifications across edges of T. Whenever the underlying walk immediately crosses an edge and returns, the two steps cancel, because the edge maps are injective. Reducing the walk proves that a vertex of one X_v cannot be identified with a different vertex of the same X_v. It also proves that the set $S(x)$ of vertices v for which x has a representative in X_v is a connected subtree: representatives along the unique path between any two such vertices are forced and exist.

If two global vertices x, y are represented in both X_u and X_v, they are represented in every block along the path from u to v. If they are adjacent in one of these blocks, fullness of the first edge image puts their edge in that edge complex, and hence in the next block. Repeating this argument transports adjacency along the path. More generally, the identical argument transports every finite simplex. Consequently every block is full in the union.

Now let $x_1, \dots, x_m$ be pairwise adjacent vertices of X. Their support subtrees $S(x_i)$ meet pairwise, since every edge of X comes from a block. Finite families of pairwise intersecting subtrees of a tree have a common vertex. Here is an explicit proof of that fact. Root T at any vertex. In each nonempty subtree take the unique vertex nearest the root, and choose one, p, of greatest distance among these finitely many nearest vertices. For another subtree choose a point where it meets the subtree associated to p. The two nearest vertices lie on the rooted path to that point; the deeper one is p. Connectedness of the other subtree therefore puts p in it. Thus p lies in every subtree.

All x_i are consequently represented in X_p. Adjacency transports to that block as above. Since X_p is flag, these representatives span a simplex in X_p and therefore in X. □

If Y is flag and σ is a simplex, its closed star is a full subcomplex: its vertices are those of σ and all vertices adjacent to every vertex of σ. Any simplex on these vertices joins σ to a clique, and the flag property fills that clique. This observation supplies the full edge inclusions in the next result.

Theorem 6. *Let $\alpha = (A, (A_i)_{i \in I_A})$ and $\beta = (B, (B_i)_{i \in I_B})$ be flag-transitive coset geometries of finite rank. Write $L = I_A \cap I_B$, $P = I_A \setminus L$ and $Q = I_B \setminus L$. Suppose a fixed isomorphism identifies $C = A_P$ with B_Q and identifies $A_{P \cup J}$ with $B_{Q \cup J}$ for every $J \subseteq L$. In $G = A *_C B$ put*

$$G_i = \langle A_{\{i\} \cap I_A}, B_{\{i\} \cap I_B} \rangle \quad (i \in I_A \cup I_B).$$

Then the coset incidence system $(G,(G_i))$ is a flag-transitive geometry. Its chamber stabilizer is the common subgroup $H = A_{I_A} = B_{I_B}$, and for every $J \subseteq I_A \cup I_B$,

$$G_J = A_{J\cap I_A} *_{C\cap A_{J\cap I_A}} B_{J\cap I_B}.$$

Proof. We use the usual tree of the amalgam: vertices are left cosets gA, gB, edges are gC, and gC joins gA to gB. The reduced word theorem proves that this graph is a tree. Using left cosets here only changes the convention by inversion from the usual right-coset geometry.

At an A vertex put the flag complex of α joined to a simplex whose types are Q. At a B vertex put the flag complex of β joined to a simplex of types P. At an edge put the flag complex of the common residue

$$(C,(C\cap A_l)_{l\in L}) = (C,(C\cap B_l)_{l\in L})$$

joined to the simplex of types $P\cup Q$. Flag transitivity identifies this residue with the link of the standard P face in α and with the link of the standard Q face in β. In each endpoint block the edge complex is therefore the closed star of the corresponding face, with the extra simplex already adjoined. These embeddings are full. The construction is transported equivariantly to every coset vertex and edge. Lemma 5 gives a flag complex X.

Every block chamber has one vertex of each type in $I_A \cup I_B$. Flag transitivity in the two input geometries identifies its chambers with the appropriate cosets of H. On an edge, the same identification is by C/H, so these labels agree under every gluing. Thus the chambers of X are naturally G/H: equality of chamber labels is preserved along the unique path of blocks supporting the chamber. In particular G acts transitively on them and their stabilizers are conjugates of H.

We give the same argument for an arbitrary standard face of type J, including the empty face. Its stabilizer in the A block is $D_A = A_{J\cap I_A}$, its stabilizer in the B block is $D_B = B_{J\cap I_B}$, and its stabilizer on the edge is $D_C = C\cap D_A = C\cap D_B$. The equality uses the fixed compatible identification, not merely an abstract isomorphism of these groups. The copies of this face that are identified across edges are exactly the orbit of the standard copy under $\langle D_A, D_B\rangle$. To justify this assertion locally, an edge aC at the A vertex contains the standard J face precisely when its P face aP is incident with the standard $J\cap I_A$ face. Their union is then a flag. Input flag transitivity supplies $d \in D_A$ carrying the standard P face to aP, so $aC = dC$. Thus the supporting incident edges are exactly $D_A C/C$. At the other endpoint they are exactly $D_B C/C$. A path transporting the face crosses only these edge copies, so its consecutive changes of representatives are in D_A or D_B. Conversely each such change fixes the face. The reduced word theorem identifies this generated subgroup with $D_A *_{D_C} D_B$.

Taking a one-vertex face identifies its stabilizer with the stated G_i. Thus vertices of X are precisely the corresponding cosets. Two vertices in a simplex have a common chamber and hence a common coset representative; conversely a common representative places them in a translate of the standard chamber. Their incidence is therefore exactly coset incidence. Taking the whole standard J face gives the displayed intersection formula, because its stabilizer is the intersection of its vertex stabilizers. Finally every simplex of X is contained in a block chamber, so all flags extend to chambers. This proves the geometric assertion. □

Corollary 7. *Under the hypotheses of Theorem 6, the amalgam geometry is residually connected if and only if both input geometries are. It is firm, thin or thick if and only if both inputs have the respective property. Here the group actions need not be faithful.*

Proof. The residue of a standard face of type J has group G_J and parabolics $G_{J\cup\{i\}}$, for $i \notin J$. The intersection formula therefore describes it by the same construction, using the corresponding residues of the two input geometries. If $P \subseteq J$, its A factor is already in C and the amalgam collapses to the B factor. The residue is exactly the residue of β at $J\cap I_B$. If $Q \subseteq J$, the symmetric statement holds. In particular, fixing the standard Q face recovers α, and fixing the standard P face recovers β; an empty such face simply means the whole geometry. This proves necessity of residual connectivity.

For sufficiency, take a residue with at least two types left. The cases just described reduce to residues of the inputs. Otherwise both $P \setminus J$ and $Q \setminus J$ are nonempty. In the tree construction of this residue, each A block is a cone on its nonempty remaining Q simplex, and each B block is a cone on its nonempty remaining P simplex. Each block is connected. The edge complexes contain the nonempty simplex on $(P \cup Q) \setminus J$, so adjacent blocks meet. Their union over a connected tree is connected. This includes all residues of rank at least two and proves residual connectivity.

For a panel, $J = I \setminus \{i\}$ necessarily contains P or Q. The panel is consequently exactly an input panel by the first paragraph. If $i \in L$, the two possible descriptions agree under the fixed identification in C. Thus the panel sizes are the original indices $[A_{I_A \setminus \{i\}} : H]$ or $[B_{I_B \setminus \{i\}} : H]$, with every original index represented. Comparison with 1, 2 or 3 gives the three assertions. □

Converse for a compatible amalgam

The converse uses a single-subgroup embedding, which is valid without the false unrestricted intersection formula for pairs of special subgroups. Let I_A, I_B be arbitrary finite type sets, put $L = I_A \cap I_B$, $P = I_A \setminus L$, $Q = I_B \setminus L$, and suppose a fixed isomorphism identifies $C = A_P$ with B_Q and identifies $A_{P \cup J}$ with $B_{Q \cup J}$ for every $J \subseteq L$. No flag assumption is imposed initially. Set $G = A *_C B$ and $G_i = \langle A_{\{i\} \cap I_A}, B_{\{i\} \cap I_B} \rangle$.

Proposition 8. *If $(G, (G_i))$ is flag-transitive, both input incidence systems are flag-transitive. Together with Theorem 6, this is an if and only if statement.*

Proof. For $q \in Q$, we have $G_q = \langle A, B_q \rangle$ and $C \leq B_q$. Fix left transversals for C in A and B. A reduced amalgam word belongs to $\langle A, B_q \rangle$ precisely when all its B syllables belong to B_q. This criterion is independent of the choices of representatives, since multiplying a syllable by an element of C preserves membership in B_q. It follows from uniqueness of the amalgam normal form, or by choosing the B transversal to extend one for C in B_q. Intersecting over $q \in Q$, every such syllable must belong to $\bigcap_{q \in Q} B_q = C$, so there are no nontrivial B syllables. Hence $G_Q = A$. If Q is empty then $C = B$ and $G = A$, so the same equality holds. Symmetrically $G_P = B$.

For $i \in I_A$, the two generating subgroups of G_i have the same intersection with C, by the fixed compatibility assumption. The reduced word theorem embeds their amalgam in G and gives $G_i \cap A = A_i$. This is the single-subgroup embedding statement; it does not use the false unrestricted intersection formula for two special amalgam subgroups. In the flag-transitive geometry of G, the residue of its standard Q face is therefore the coset geometry with group $G_Q = A$ and parabolics $G_Q \cap G_i = A_i$, $i \in I_A$. Every residue of a flag-transitive geometry is flag-transitive. This proves the assertion for α, and the standard P residue gives β. Empty private type sets cause no exception: the corresponding residue is the whole geometry. □

2.1 Free products and their panel indices

Lemma 9. *Let $\mathcal{P}$ and $\mathcal{Q}$ be families of subgroups of A and B, respectively, such that their left cosets are Helly. Then the left cosets of the subgroups $P * Q \leq A * B$, with $P \in \mathcal{P}$ and $Q \in \mathcal{Q}$, are Helly. Moreover,*

$$\bigcap_{j=1}^{m} (P_j * Q_j) = \left(\bigcap_{j=1}^{m} P_j \right) * \left(\bigcap_{j=1}^{m} Q_j \right).$$

Proof. The intersection assertion follows from the unique reduced word in a free product: a word belongs to $P * Q$ exactly when each of its A syllables belongs to P and each of its B syllables belongs to Q. This also proves $(P * Q) \cap A = P$ and $(P * Q) \cap B = Q$.

Consider the tree with vertices gA, gB and with an edge labelled g joining them, for every $g \in A * B$. For $H = P * Q$, the union of the edges labelled by elements of H is a connected subtree. Indeed a reduced word in P and Q gives a path of such edges from the identity edge to its own edge. Conversely

the inclusions of the two factors make this the usual tree of $P * Q$ embedded in the ambient tree. Denote the translate associated to the coset gH by $T(gH)$. Its edge labels are exactly the elements of gH.

Take finitely many pairwise intersecting cosets $g_j H_j$. Their subtrees meet pairwise in edges and hence have a common vertex v, by the finite Helly property for subtrees of a tree. Suppose $v = xA$; the other case is identical. Each set $g_j H_j \cap xA$ is nonempty and, after multiplication by x^{-1}, is a left coset of P_j in A. For two of our subtrees, choose an edge whose label belongs to both original cosets. The path from that edge to v lies in both subtrees. Its edge incident with v has a single label belonging to both cosets and to xA. Thus these local P_j cosets meet pairwise. The assumed Helly property in A gives a common element, which belongs to every original coset. The tree Helly fact follows, for example, by rooting the tree and choosing the deepest of the nearest-to-root points of the finitely many subtrees. □

Theorem 10. *Let $\alpha = (A, (A_i)_{i \in I_A})$ and $\beta = (B, (B_i)_{i \in I_B})$ be coset incidence systems with disjoint type sets. In $G = A * B$ put $G_i = A_i * B$ for $i \in I_A$ and $G_i = A * B_i$ for $i \in I_B$. Then*

$$G_J = A_{J \cap I_A} * B_{J \cap I_B}.$$

The resulting system is a flag-transitive geometry if and only if both input systems are flag-transitive. Under these conditions it is residually connected if and only if both inputs are.

Proof. The displayed formula and forward flag assertion are Lemmas 1 and 9, applied to the families of all input parabolics. For necessity, lift a flag of α to the corresponding G_i cosets using the same representatives in A. Pairwise intersections remain nonempty. A common element in G, followed by the retraction $G \to A$ that kills B, gives a common element of the original cosets, since G_i maps onto A_i for $i \in I_A$. This proves flag transitivity of α without any Borel assumption. The retraction to B gives the other input.

For a flag-transitive geometry, connectedness of a residue of rank at least two is equivalent to generation of its group by the maximal parabolics in that residue. To see this criterion, chambers incident along an element connect their representatives through the appropriate parabolic coset; the component containing the standard chamber consists exactly of the chambers represented by the generated subgroup. Flag transitivity identifies every residue with its parabolic coset system.

Apply this criterion to a face of type J. If both $I_A \setminus J$ and $I_B \setminus J$ are nonempty, a remaining A type has residue parabolic containing the whole factor $B_{J \cap I_B}$, and a remaining B type has one containing $A_{J \cap I_A}$. These two factors generate G_J. If only A types remain, there are at least two of them. The residue parabolics generate $G_J = A_{J \cap I_A} * B_{I_B}$ exactly when their projections generate $A_{J \cap I_A}$, which is the required residue condition in α. The case with only B types is symmetric. Taking $J = J_A \cup I_B$, or $J = I_A \cup J_B$, proves necessity for every input residue as well. □

Proposition 11. *Put $H_A = A_{I_A}$ and $H_B = B_{I_B}$. For a type $i \in I_A$, write $U = A_{I_A \setminus \{i\}}$ and $U_0 = H_A$. The panel size in the free-product geometry is*

$$[U * H_B : U_0 * H_B] = \begin{cases} 1, & U = U_0, \\ [U : U_0], & H_B = 1, \\ \infty, & U_0 < U \text{ and } H_B \neq 1. \end{cases}$$

There is the symmetric formula for a type in I_B. Consequently the free-product geometry is firm if and only if both inputs are firm. If both type sets are nonempty, it is thin if and only if both inputs are thin and $H_A = H_B = 1$.

Proof. The Borel is $H_A * H_B$ by Theorem 10, giving the index. The first two cases follow directly. For the last, choose $u \in U \setminus U_0$ and $1 \neq v \in H_B$. The cosets $(uv)^n (U_0 * H_B)$, $n \geq 0$, are distinct: equality would put a nonzero power of uv in $U_0 * H_B$, contrary to its reduced word. This proves infinitude, including when U or H_B is infinite. An input panel has size one precisely when $U = U_0$, which proves

firmness. Under thinness all input panels must be nontrivial. If both type sets occur, the last case then forces both Borels to be trivial; the second case recovers exactly the input panel sizes. The converse is immediate from the same formulas. □

3 HNN extensions and indexed parabolics

Let A be a group and let $\mathcal{P}$ be a family of its subgroups, closed under finite intersections. We say that $\mathcal{P}$ has the *coset Helly property* if every finite family of pairwise intersecting left cosets of members of $\mathcal{P}$ has nonempty total intersection. We include A in $\mathcal{P}$. By Lemma 1, all standard parabolics of a finite-rank flag-transitive coset geometry form such a family.

Fix $C, D \in \mathcal{P}$ and an isomorphism $\phi : C \longrightarrow D$. Write

$$G = \langle A, t \mid t^{-1}ct = \phi(c)\ (c \in C)\rangle.$$

A subgroup $P \in \mathcal{P}$ is called compatible if $\phi(P \cap C) = P \cap D$. Put $\widehat{P} = \langle P, t\rangle$. Britton's lemma embeds the HNN extension of P along these intersections in G and gives $\widehat{P} \cap A = P$. These are retained classical normal-form facts. Unlike an arbitrary family of special subgroups, the family below keeps its local coset Helly hypothesis.

Theorem 12. *If $\mathcal{P}$ has the coset Helly property, then the family consisting of all $P \in \mathcal{P}$ and all $\widehat{P}$ for compatible $P \in \mathcal{P}$ has the coset Helly property in G. For compatible P, Q,*

$$\widehat{P} \cap \widehat{Q} = \widehat{P \cap Q}, \qquad R \cap \widehat{P} = R \cap P \quad (R \in \mathcal{P}).$$

Proof. Use the Bass–Serre tree T of this HNN extension. Its vertices are gA and its oriented edges are gC, with endpoints gA and gtA. The latter endpoint is independent of the representative because $ct = t\phi(c)$. At a vertex xA, outgoing edge directions are parametrized by left cosets of C in A, and incoming directions by left cosets of D in A.

For compatible P, the embedded Bass–Serre tree of $\widehat{P}$ has vertices hA and edges hC, $h \in \widehat{P}$. Its injectivity follows from the same reduced word theorem, since the associated intersections are exactly $P \cap C$ and $P \cap D$. Denote this connected subtree by T_P. For a base subgroup $R \leq A$ instead use the singleton subtree $\{A\}$. Attach to a coset $g\widehat{P}$ the subtree gT_P and to a coset gR the vertex gA. These assignments are independent of the chosen coset representative.

If a vertex $v = xA$ belongs to such a support subtree, the corresponding coset meets xA. After multiplication by x^{-1} this intersection is a left coset in A of P, or of R in the base-subgroup case. For example, choose $y \in g\widehat{P} \cap xA$; the intersection is $y(\widehat{P} \cap A) = yP$. An incident edge belongs to gT_P precisely when this local P coset meets the local C or D coset describing the direction of that edge. This follows directly from the vertex and edge descriptions of the embedded tree.

We need one transport observation. Suppose two global cosets meet, and a vertex v lies in both support subtrees. Choose a common element z of the cosets. Both support subtrees contain zA, so they contain the whole finite path from zA to v. Start with the common representative z at its first vertex. At the next edge, the two local subgroup cosets intersect each other and each meets the local associated-subgroup coset of that edge. Local coset Helly supplies a common element of all three. For an outgoing edge multiply this representative on the right by t; for an incoming edge multiply it by t^{-1}. The result lies at the next vertex and still belongs to both global cosets, because both subgroups contain t. If either subgroup was a base subgroup, its support was a singleton and the path has length zero, so no multiplication is needed. Repeating the step gives a common representative in xA.

Now take finitely many pairwise intersecting global cosets. Their support subtrees intersect pairwise. The elementary Helly property for subtrees gives a common vertex $v = xA$. By the transport observation every pair of local fibers at v intersects. These fibers are cosets of members of $\mathcal{P}$, so local coset Helly supplies an element in all of them. Multiplying by x proves the first assertion.

For the intersection formula, first observe that $P \cap Q$ is compatible: an isomorphism commutes with intersections. Thus $K = \widehat{P \cap Q}$ is contained in both groups. Take $g \in \widehat{P} \cap \widehat{Q}$ and follow the finite path from A to gA. Both embedded subtrees contain it. We show inductively that every vertex on this path is kA for some $k \in K$. This holds initially with $k = 1$. Translate a current vertex kA and its next edge by k^{-1}. If that edge is outgoing, its local coset aC meets both P and Q; if incoming, its local coset aD does. Since $P \cap Q$ contains 1, the three local cosets meet pairwise. The local Helly property supplies $p \in P \cap Q$ in the relevant coset. The next vertex is therefore $kptA$ or $kpt^{-1}A$, respectively, and its representative is again in K. At the end $gA = kA$, and hence $k^{-1}g \in A \cap \widehat{P} \cap \widehat{Q} = P \cap Q \leq K$. This proves the first equality. The second follows immediately from $\widehat{P} \cap A = P$. □

Corollary 13. *Let $(A, (A_i)_{i\in I})$ be a flag-transitive coset geometry of finite rank. Let C, D be standard parabolics, and let $\phi : C \to D$ be an isomorphism. Choose a partition $\mathcal{S}$ of I such that every $A_s = \bigcap_{i\in s} A_i$, $s \in \mathcal{S}$, is compatible, and choose $F \subseteq I$. In the HNN extension take maximal parabolics*

$$G_s = \langle A_s, t\rangle \quad (s \in \mathcal{S}), \qquad G_* = A_F.$$

This is a flag-transitive coset geometry. For a set J of its types, write $E = \bigcup_{s\in J\cap\mathcal{S}} s$. Then

$$G_J = \begin{cases} \langle A_E, t\rangle, & * \notin J, \\ A_{E\cup F}, & * \in J. \end{cases}$$

Its chamber stabilizer is A_I.

Proof. Take $\mathcal{P}$ to be all standard parabolics of the input, including A. These have the local coset Helly property explained above. Theorem 12 applies to all the displayed maximal parabolics. A flag is a finite family of pairwise intersecting such cosets, so it has a common representative. It is therefore a translate of a standard flag and extends to a standard chamber. The intersection formulas follow by induction from the theorem, with the usual empty-intersection convention. For the complete type set $E = I$, so the Borel is A_I. □

Corollary 14. *Suppose the input in Corollary 13 is residually connected. Let $C = A_J$, $D = A_K$, and suppose a specified bijection $f : I \setminus J \longrightarrow I \setminus K$ satisfies*

$$\phi(A_{J\cup X}) = A_{K\cup f(X)} \qquad (X \subseteq I \setminus J).$$

Take $\mathcal{S}$ to be the connected classes of the arrows $i \to f(i)$ on I, and put $F = J \setminus K$. Then the resulting HNN geometry is residually connected. Its $*$ panels are infinite. Its panel of type $s \in \mathcal{S}$ has size

$$[A_{I\setminus(s\setminus F)} : A_I].$$

In particular it is firm when the input is firm.

Proof. Write $H = A_I$ and $R_i = A_{I\setminus\{i\}}$. The displayed compatibility gives $\phi(H) = H$ and $\phi(R_i) = R_{f(i)}$ for $i \notin J$. Each class s satisfies $f(s \setminus J) = s \setminus K$. Consequently $\phi(A_s \cap C) = A_s \cap D$, so the previous corollary applies. Its minimal parabolics are

$$M_s = A_{I\setminus(s\setminus F)}, \qquad M_* = \langle H, t\rangle.$$

Residual connectivity of the input gives the elementary generation rule

$$A_{I\setminus E} = \langle H, R_i \ (i \in E)\rangle \quad (E \subseteq I).$$

This is Lemma 4, including the empty support; the Borel is not dropped.

Let W be a subset of the new type set $\mathcal{S} \cup \{*\}$, and put $E = \bigcup_{s\in W\cap\mathcal{S}} s$. If $* \notin W$, then

$$\langle H, M_s \ (s \in W)\rangle = A_{I\setminus(E\setminus F)} = G_{(\mathcal{S}\cup\{*\})\setminus W}.$$

If $* \in W$, this generated group also contains t. Every $i \in E \cap F$ is in the range but not the domain of f. It therefore has a predecessor $j \in E \setminus F$ with $f(j) = i$. Conjugation by t brings R_i into the group from R_j. Thus in this case the generated group is $\langle H, t, R_i \ (i \in E)\rangle = \langle A_{I\setminus E}, t\rangle$, again exactly $G_{(\mathcal{S}\cup\{*\})\setminus W}$.

We have proved the same generation rule for every new standard parabolic. If at least two types remain, each of its minimal generators belongs to one of the parabolics obtained by deleting a different remaining type. Those parabolics therefore generate it. The residual generation criterion proves residual connectivity.

Since $\phi(H) = H$, the subgroup $\langle H, t\rangle$ is an extension of H by the infinite cyclic group generated by the stable letter. Its index over H is infinite. The other panel indices are the stated indices of M_s over H. Every class has a point outside F: a point in F has a predecessor outside F. If the input is firm, the group M_s therefore contains some R_i strictly larger than H. All these panels have at least two elements, as required. □

3.1 Firmness recovers the missing indices

Let $\alpha = (A, (A_i)_{i\in I})$ be a coset geometry of finite rank. Write $A_J = \bigcap_{j\in J} A_j$, $A_\varnothing = A$, $H = A_I$, and $R_i = A_{I\setminus\{i\}}$. The following criterion turns an isomorphism of unlabelled parabolic lattices into the indexed compatibility needed above. Firmness is an actual hypothesis.

Lemma 15. *Suppose $R_i > H$ for every $i \in I$. Then*

$$A_J \le A_K \quad \Longleftrightarrow \quad K \subseteq J.$$

Consequently the lattice of standard parabolics, ordered by inclusion, is the reverse Boolean lattice on I.

Proof. If $K \subseteq J$, the inclusion follows from the definition of an intersection. Conversely, suppose $k \in K \setminus J$. Then $R_k \le A_J$, because every member of J differs from k. If $A_J \le A_K$, it follows that $R_k \le A_k$, whence $R_k = R_k \cap A_k = H$, a contradiction. This proves the equivalence, including injectivity of $J \mapsto A_J$. □

Corollary 16. *Suppose α is flag-transitive and firm. Let $\phi : A_J \longrightarrow A_K$ be an isomorphism inducing an isomorphism of the unlabelled lattices of standard parabolics contained in its domain and codomain. Then there is a unique bijection $f : I \setminus J \longrightarrow I \setminus K$ such that*

$$\phi(A_{J\cup X}) = A_{K\cup f(X)} \qquad (X \subseteq I \setminus J).$$

Thus the indexed hypothesis of Corollary 14 follows from the source's admissibility condition for firm flag-transitive inputs.

Proof. Flag transitivity identifies the chambers on a standard panel of missing type i with R_i/H. Firmness therefore implies $[R_i : H] \ge 2$. Lemma 15 applies. The standard parabolics contained in A_J are precisely the $A_{J\cup X}$, and their coatoms are $A_{J\cup\{i\}}$ for $i \notin J$. The isomorphism permutes coatoms bijectively, giving f. Since it preserves intersections, intersecting these coatom equalities proves the formula for every X, including $X = \varnothing$.

For completeness, form the equivalence classes s of the arrows $i \mapsto f(i)$ on I. Bijection of the domain and range gives $f(s \setminus J) = s \setminus K$. Hence

$$\phi(A_s \cap A_J) = \bigcap_{i\in s\setminus J} \phi(A_{J\cup\{i\}}) = \bigcap_{j\in s\setminus K} A_{K\cup\{j\}} = A_s \cap A_K.$$

Both displayed intersections of images are taken inside A_K; if their index sets are empty they mean A_K, in agreement with $\phi(A_J) = A_K$. Every A_s is therefore compatible. These are precisely the subgroups required by the local coset Helly theorem. Its flag-transitivity conclusion follows; its residual-connectivity conclusion also requires residual connectivity of the input, as stated there. □

3.2 Residually connected inputs with universal types

Let $\alpha = (A, (A_i)_{i\in I})$ be a finite-rank flag-transitive, residually connected coset geometry. Write $A_J = \bigcap_{i\in J} A_i$, $H = A_I$, and $R_i = A_{I\setminus\{i\}}$. The Borel-preserving residual generation formula is

$$A_J = \langle H, R_i \,:\, i \notin J\rangle. \tag{6}$$

This is the same Borel-preserving formula proved in Lemma 4.

Lemma 17. *Put $I_0 = \{i \,:\, A_i = A\}$ and $I_+ = I \setminus I_0$. Then $A_i = A$ if and only if $R_i = H$. Deleting the universal types I_0 leaves a firm flag-transitive, residually connected geometry on I_+ with the same unlabelled family of standard parabolics and the same Borel H. The original incidence geometry is its join with one singleton for each type in I_0.*

Proof. If $A_i = A$, then $H = A_i \cap R_i = R_i$. Conversely, if $R_i = H$, formula (6) for $J = \varnothing$ and $J = \{i\}$ gives $A = A_i$. A type with maximal parabolic A has exactly one element, incident with every other element, proving the join description. Dropping these factors leaves every standard intersection unchanged. Flags still have common representatives. Every residue on at least two active types is a residue of the original geometry after fixing the deleted universal elements, so it remains connected. Each active panel has index $[R_i : H] > 1$, proving firmness. The empty active geometry is interpreted as a single empty chamber. □

For later comparison we also make the firm branch explicit without assuming residual connectedness. For a firm flag-transitive input and a source-admissible $\phi : A_J \to A_K$, Lemma 15 gives the indexed partial bijection required by Theorem 12. For its classes s and $F = J \setminus K$, that theorem's intersection formula gives the Borel H, the stable minimal parabolic $\langle H, t\rangle$, and the other minimal parabolics

$$M_s = A_{I\setminus(s\setminus F)}.$$

The first has infinite index over H, since the stable letter has infinite image in $\mathbb{Z}$ and normalizes H. Every class has an element outside F: a point in F is in the range but not the domain of the partial bijection, and its predecessor is outside J, hence outside F. Consequently M_s contains some $R_i > H$. All panels are firm. This proves the full firm-input branch of source Theorem 3.35, including part (c), independently of the residual-connectivity assertion below.

Suppose $C = A_J, D = A_K$, and $\phi : C \to D$ satisfies the source's Definition 3.30: it induces an isomorphism of the unlabelled families of standard parabolics lying in these subgroups. The source forms classes of the arrows $i \to j$, with $i \notin J$, $j \notin K$, for which $\phi(A_{J\cup\{i\}}) = A_{K\cup\{j\}}$. It uses $F = J \setminus K$, the maximal subgroups $\langle A_s, t\rangle$ for these connected classes s, and the stable-type subgroup A_F in the HNN extension $\langle A, t \mid t^{-1}ct = \phi(c)\rangle$.

Theorem 18. *With these literal choices, every flag-transitive, residually connected input gives a flag-transitive, residually connected output. Thus the residually connected branch of Theorem 3.35 remains valid without firmness. Active output panels have the indices in the indexed HNN theorem; each additional universal class has one element in its panel.*

Proof. Apply Lemma 17. On the active types the firm geometry has the reverse Boolean parabolic lattice proved in Lemma 15. Thus the isomorphism ϕ gives a bijection

$$f : I_+ \setminus J \longrightarrow I_+ \setminus K$$

with $\phi(A_{J\cup X}) = A_{K\cup f(X)}$ for every $X \subseteq I_+ \setminus J$. Here intersections with J, K can ignore their inactive entries. This follows because the coatoms of the Boolean interval inside A_J are exactly the subgroups $A_{J\cup\{i\}}$ for its missing active types; an isomorphism permutes those coatoms and commutes with all their intersections. The minimal subgroup H is also preserved.

For an active $i \notin J$, the subgroup $A_{J\cup\{i\}}$ is proper in A_J, whereas for an inactive i it equals A_J. The same distinction holds on the K side. Hence a source arrow never connects an active type to an inactive one. On active types the connected classes are exactly those of the partial bijection f, with isolated active types included. The stable parabolic is $A_F = A_{F\cap I_+}$. Thus all active-class parabolics and the stable parabolic coincide exactly with the indexed HNN construction of Corollary 14 on the firm active geometry. Its flag transitivity and residual connectedness follow from that theorem.

An inactive class s consists only of types with $A_i = A$, so $A_s = A$ and $\langle A_s, t\rangle$ is the entire HNN group. Its coset type is a singleton universal element. Adding any number of these types preserves flag transitivity and residual connectedness: a residue with at least two active types uses the active conclusion, a residue with one active type and at least one universal type has a connected star, and a residue consisting of at least two universal types is a complete graph. The same observation covers the case $I_+ = \varnothing$. In that case $C = D = A$ and the stable-type cosets of A in $A \rtimes \mathbb{Z}$ are the infinite rank-one part; all other types are universal. This proves the assertion and panel description. □

4 Semidirect products and orbit intersections

Let $\alpha = (A, (A_i)_{i\in S})$ and $\beta = (B, (B_j)_{j\in T})$ be flag-transitive coset geometries of finite rank. Write $H_A = A_S$ and $H_B = B_T$. Suppose that B acts on A by automorphisms and that an indexed action on S is specified, with $b(A_i) = A_{bi}$. This specification also covers coincident parabolics: an action on an unlabelled set of subgroups alone does not automatically specify an action on their repeated indices. Let $\mathcal{O}$ be the orbit partition of S.

For an orbit $s \in \mathcal{O}$, put $A_s = \bigcap_{i\in s} A_i$. In $G = A \rtimes B$ take the indexed parabolics

$$P_s = A_s \rtimes B \quad (s \in \mathcal{O}), \qquad P_j = A \rtimes B_j \quad (j \in T).$$

The orbit types here use the whole orbit intersection. The construction below instead uses orbits of parabolic subgroups; these are different incidence structures with different panel formulas.

Proposition 19. *Let $\overline{\alpha} = (A, (A_s)_{s\in\mathcal{O}})$. The coset incidence system $(G, (P_i))$ is the join $\overline{\alpha} * \beta$: the two component incidences are unchanged and every element of one component is incident with every element of the other. Both $\overline{\alpha}$ and the join are flag-transitive geometries. For $J = J_O \sqcup J_T$, put $E = \bigcup_{s\in J_O} s$. Then*

$$G_J = A_E \rtimes B_{J_T}, \qquad G_{\mathcal{O}\sqcup T} = H_A \rtimes H_B.$$

Proof. Every A_s is B-invariant. In semidirect coordinates,

$$(a, b)P_s = (aA_s) \times B, \qquad (a, b)P_j = A \times (bB_j).$$

Here the second equality uses that b acts bijectively on A. Thus an orbit-type element is determined by the coset aA_s, a T-type element by bB_j, and their incidences are exactly those stated. The parabolic formula follows by intersecting the two coordinates; this is ordinary intersection of Cartesian products.

The cosets of the A_s belong to the family of cosets of standard parabolics of α. That family is Helly by Lemma 1. Hence A is flag-transitive on $\overline{\alpha}$, and every flag there extends to a chamber. The same is assumed for β. For a flag of the join, choose $b \in B$ carrying its β part to the standard flag. This acts on its $\overline{\alpha}$ part; then an element of A carries that part to the standard flag while fixing the β part. Thus G is flag-transitive. Alternatively, a common B coordinate and a common A coordinate in the displayed cosets give a common group representative directly. □

Proposition 20. *If both inputs are residually connected, so is the semidirect join. Its panel indices are*

$$[A_{S\setminus s} : H_A] \quad (s \in \mathcal{O}), \qquad [B_{T\setminus\{j\}} : H_B] \quad (j \in T).$$

It is firm when both inputs are firm. If both inputs are thin, the join is thin exactly when every orbit in S is a singleton.

Proof. For clarity we first prove residual connectivity of $\overline{\alpha}$. Put $R_i = A_{S\setminus\{i\}}$. Residual connectivity of the finite-rank flag-transitive input gives

$$A_{S\setminus E} = \langle H_A, R_i : i \in E\rangle \qquad (E \subseteq S).$$

This is Lemma 4. Take a coarse residue with at least two orbit types remaining, whose union is E. For every $i \in E$ choose a remaining orbit s not containing i. Then $R_i \le A_{(S\setminus E)\cup s}$, one of the maximal parabolics of that residue. Those parabolics also contain H_A and therefore generate its whole group $A_{S\setminus E}$. This proves residual connectivity of the coarse geometry.

A residue of a join is the join of the corresponding component residues. If both components have a type remaining, the incidence graph is connected because all vertices across the two components are incident. If just one component has types remaining, residual connectivity is that of the corresponding input residue. This proves the first assertion, including empty component type sets.

The panel formulas follow by omitting just one type from the parabolic formula in Proposition 19. For an orbit s, choose $i \in s$. Firmness of α gives $H_A < R_i$, and $R_i \le A_{S\setminus s}$, so its panel index is at least two. The other panels have exactly their β sizes.

Suppose both inputs are thin. Singleton orbit indices are two. If s contains distinct i, k, choose $x \in R_i \setminus H_A$ and $y \in R_k \setminus H_A$. We have $R_i \cap R_k = H_A$ by their definitions. The four left cosets

$$H_A, \quad xH_A, \quad yH_A, \quad xyH_A$$

are distinct. For example, $xyH_A = xH_A$ would put y in H_A; $xyH_A = yH_A$ would put $x \in yH_Ay^{-1} \le R_k$, a contradiction; and $xyH_A = H_A$ would put $x \in R_k$. Thus $[A_{S\setminus s} : H_A] \ge 4$, ruling out thinness. The argument uses only subgroup intersections and does not assume that H_A is normal in all of A. □

4.1 A general orbit construction

Let $\alpha = (A, (A_x)_{x\in S})$ and $\beta = (B, (B_i)_{i\in I})$ be flag-transitive coset geometries. Suppose a specified homomorphism $B \to \mathrm{Aut}(A)$ and an action of B on the indexed set S satisfy $b(A_x) = A_{bx}$. The indexed action is part of the data, even if different maximal parabolics happen to coincide. Let $\mathcal{K}$ be the set of orbits on S, choose $x_L \in L$ for each $L \in \mathcal{K}$, and assume

$$B_H x_L \cap B_K x_L = B_{H\cup K} x_L \quad (H, K \subseteq I,\ L \in \mathcal{K}). \tag{7}$$

For $T = I \sqcup \mathcal{K}$, define subgroups of $G = A \rtimes B$ by

$$G_i = A_{S\setminus\bigcup_L B_i x_L} \rtimes B_i \quad (i \in I), \qquad G_L = A_L \rtimes B \quad (L \in \mathcal{K}). \tag{8}$$

Every indicated subset of S is invariant under its second factor, so these are indeed subgroups. For $J \subseteq T$ write $J_B = J \cap I$, $J_A = J \cap \mathcal{K}$, and set

$$E(J) = S \setminus \bigcup_{L\in\mathcal{K}\setminus J_A} B_{J_B} x_L.$$

Theorem 21 (Orbit construction with a Borel subgroup)**.** *The coset system in (8) is a flag-transitive geometry. All its standard parabolics, including its Borel subgroup, are*

$$G_J = A_{E(J)} \rtimes B_{J_B}, \qquad G_T = A_S \rtimes B_I. \tag{9}$$

It is finite if and only if both input geometries are finite. It is residually connected if both inputs are residually connected, and firm if both inputs are firm. Its numbers of chambers through panels are

$$[G_{T\setminus\{i\}} : G_T] = [B_{I\setminus\{i\}} : B_I] \qquad (i \in I), \tag{10}$$

$$[G_{T\setminus\{L\}} : G_T] = [A_{S\setminus B_I x_L} : A_S] \qquad (L \in \mathcal{K}). \tag{11}$$

In particular, if both inputs are thin and B_I fixes each chosen x_L, the new geometry is thin. Faithfulness of either input is not needed for this sufficient condition.

Proof. Intersections of subgroups $A_E \rtimes B_H$ are computed coordinate by coordinate. The intersection of their A-parts is obtained by taking the union of their subscripts. Different B-orbits are disjoint, and iteration of (7) says that the intersection of the old supports is the support for B_{J_B}. Intersecting with the orbit parabolics deletes precisely the orbits in J_A. Taking complements proves (9). It uses intersections of maximal parabolics, not an unjustified formula for generated supports.

To prove flag transitivity, project a flag of G-cosets to B. Old types give pairwise-intersecting maximal-parabolic cosets; orbit types project to all of B. Coset intersection from the first section gives a common $b \in B$. After left translation by $(1, b^{-1})$, each coset has a representative $(a, 1)$. Such a coset is the Cartesian product of a coset of a standard parabolic of A and the corresponding subgroup of B. Pairwise incidence therefore gives a pairwise-intersecting family of standard-parabolic A-cosets. Their common representative, again by the coset intersection lemma, realizes the entire flag. Thus every flag is an image of a standard flag and extends to a chamber. This step by itself does not require the orbit identity; that identity is needed for the exact parabolic description and the later conclusions.

Suppose both inputs are residually connected, and fix J with at least two missing types. We prove generation of G_J by the next parabolics. If no orbit type is missing, all their A-parts are A_S; (4) for B gives the required generation. If no old type is missing, choose two distinct missing orbits L, M. The two corresponding next parabolics both contain B_I. Their A-subscripts are $E(J) \cup L$ and $E(J) \cup M$, whose intersection is $E(J)$. Lemma 4 gives the entire A-part.

In the remaining case choose a missing old type i and a missing orbit type L. The parabolic $G_{J\cup\{L\}}$ contains the full B_{J_B}. Write $E' = E(J\cup\{i\})$. The group generated by these two next parabolics therefore contains $A_{bE'}$ for every $b \in B_{J_B}$. In each remaining orbit, $B_{J_B\cup\{i\}}x_M$ is nonempty and contains x_M. Its B_{J_B}-saturation is exactly $B_{J_B}x_M$. Consequently

$$\bigcap_{b\in B_{J_B}} bE' = E(J).$$

There are only finitely many distinct subsets of the finite set S, so Lemma 4 applies directly to these conjugate parabolics. They generate $A_{E(J)}$, proving the remaining case. The residue identification and connectivity criterion already proved now establish residual connectedness in every residue.

Putting $J = T \setminus \{t\}$ in (9) gives the two panel formulas. For an old type the A-parts agree; for an orbit type the B-parts agree. These observations prove the stated indices even when the Borel subgroups are nontrivial. If the inputs are firm, the old indices are at least two. The orbit $B_I x_L$ contains x_L, so $A_{S\setminus\{x_L\}} \le A_{S\setminus B_I x_L}$; its index over A_S is at least two by firmness of α. This proves firmness. If the inputs are thin and $B_I x_L = \{x_L\}$, all the displayed indices are two, proving the sufficient thinness assertion.

Finally the number of chambers is the index of G_T. The elementary coset index formula gives

$$[G : G_T] = [A : A_S][B : B_I],$$

with the equality interpreted as a product of cardinalities. For a finite number of types, a geometry has finitely many elements if and only if it has finitely many chambers: every element extends to a chamber and a chamber is a tuple of one element of each type. The same argument applies to each input. Thus the product is finite exactly when both input geometries are finite. No finiteness of the possibly unfaithful groups A and B was assumed. □

Corollary 22 (Faithful regular inputs)**.** *For faithful regular hypertopes α, β, the construction is a regular hypertope with full type-preserving automorphism group $A \rtimes B$. Writing $A(R) = \langle a_x : x \in R\rangle$, its parabolics are*

$$G_J = A\left(\bigcup_{L\not\in J_A} B_{J_B}x_L\right) \rtimes B_{J_B}. \tag{12}$$

Its distinguished involutions are the b_i and the a_{x_L}.

Proof. The faithful regular model has $A_S = B_I = 1$ and $A_E = A(S \setminus E)$, as proved in the first section. The preceding theorem supplies flag transitivity, residual connectedness and thinness. Its Borel subgroup is trivial, and the panel groups are precisely the cyclic groups on the indicated involutions. Chamber propagation shows that the displayed regular action is the full type-preserving group. □

This recovers the general finite-type twisting theorem and its regular specialization in Piedade–Tranchida [2, Theorem 3.16 and Corollary 3.17]. The exact panel formula also explains why passing to a faithful quotient of B cannot silently preserve an arbitrary action on the types of α: its Borel subgroup must act trivially for that action to descend.

5 Invariant quotients and orbital presentations

Lemma 23 (Quotients)**.** *Use the faithful regular input geometries and specified indexed action of Corollary 22. If $M \leq A$ is invariant under B, its normal closure N in A is invariant. The action descends to A/N and permutes the images of the standard maximal parabolics with the same indexed type action. If their coset geometry is a regular hypertope, the corresponding twisting is again a regular hypertope.*

Proof. The normal closure is generated by ama^{-1} with $a \in A, m \in M$. An automorphism from B carries each such generator to another of the same form, and its inverse gives equality. The quotient action and parabolic permutation follow. Orbit identity (7) concerns only the indexed types and is unchanged.

For completeness, the regular-geometry assumption can be interpreted without silently assuming a faithful quotient action. Put $\bar{A} = A/N$, $C_x = A_x N/N$, and $C_E = \bigcap_{x \in E} C_x$, with $C_\emptyset = \bar{A}$. The source's quotient twisting uses intersections of these maximal parabolics:

$$\bar{G}_i = C_{S \setminus \bigcup_L B_i x_L} \rtimes B_i \quad (i \in I), \qquad \bar{G}_L = C_L \rtimes B \quad (L \in \mathcal{K}). \tag{13}$$

These are not defined by substituting the images of generated-support subgroups into (12).

Each image of a_x fixes the base flag with type x deleted. It cannot belong to C_x: otherwise C_x, already containing the other generators, would be all of $\bar{A}$, giving only one element of type x, contrary to thinness. It therefore performs the unique adjacency of that type. Chamber-graph connectivity makes $\bar{A}$ flag-transitive. Its chamber stabilizer $H = C_S$ fixes every chamber by thinness, so is its normal action kernel. It is B-invariant because it is the intersection of the permuted C_x.

Every C_E contains H, and intersections commute with passage to $\bar{A}/H$. In that faithful regular model formula (1) therefore gives

$$C_E/H = \bigcap_{x \in E} (C_x/H) = \langle a_x H : x \in S \setminus E \rangle.$$

Consequently every subgroup in (13) is the full inverse image of the corresponding subgroup from Corollary 22 for $\bar{A}/H$. Apply that theorem. Passage through the normal subgroup $H \rtimes 1$ preserves all cosets and their intersections, so the original quotient twisting has the same regular geometry. For a transitive type orbit $L = S$, its new parabolic is $H \rtimes B$, not necessarily B. The quotient group $\bar{A}$ itself is not asserted to be the faithful automorphism group. □

Proposition 24 (Orbital presentation)**.** *Let B act transitively on a finite set X with base point x_0 and stabilizer P. Give each orbit of unordered distinct pairs a Coxeter label. Let Y be a finite set fixed by B, with fixed labels within Y and one label m_y between y and every element of X. Put $A = W(Y \sqcup X)$. Then $A \rtimes B$ has the presentation consisting of a presentation for B, one involution $a = a_{x_0}$, the involutions c_y for $y \in Y$, and:*

1. *$[a, p] = 1$ for generators p of P, and $[c_y, b] = 1$ for generators b of B;*

2. *the prescribed relations among the c_y, and $(c_y a)^{m_y} = 1$ when $m_y < \infty$;*

3. $(a\tau a\tau^{-1})^m = 1$ *for one representative $\{x_0, \tau x_0\}$ of each unordered pair orbit of finite label m.*

The unordered orbits are indexed by the inverse-paired double cosets $P\tau P \cup P\tau^{-1}P$ with $\tau \notin P$.

Proof. In the proposed group define $a_{bx_0} = bab^{-1}$. The stabilizer relations make this independent of b. Double-coset multiplication changes the representative without changing its pair orbit, and inversion interchanges its endpoints. Conversely, a transporter of two unordered pairs either fixes the selected first endpoint or interchanges the endpoints, giving exactly the two double cosets displayed. Hence conjugates of the chosen relations give every Coxeter relation on X. Conjugating the relations with c_y, which is fixed by B, gives all the mixed Coxeter relations. These elements therefore define a homomorphism from $W(Y \sqcup X) \rtimes B$ onto the proposed group. The reverse map takes the proposed generators to their named elements in the semidirect product; all the relators hold there. Both composites fix the generating set, proving the isomorphism. This is a presentation of the full group, not merely a finite image. □

Lemma 25 (Exact doubled label)**.** *In $A \rtimes B$ let $a = a_x$ and let $b \in B$ be an involution with $bx \neq x$. If $a_x a_{bx}$ has order m (possibly infinite), then ab has order $2m$, with $2\infty = \infty$.*

Proof. We have $(ab)^2 = a_x a_{bx}$. Projection to B sends ab to the nontrivial involution b, so any power of ab equal to 1 has even exponent. For an even exponent $2k$ it is 1 precisely when $(a_x a_{bx})^k = 1$. This proves exactness in both the finite and infinite cases. □

5.1 Indexed quotient actions without a geometric assumption

For a group A with a specified action of B, a B-invariant subgroup H has invariant normal closure N: every image of $a^{-1}ha$ is a conjugate of an element of H, and applying the inverse automorphism gives equality. For arbitrary H, the subgroup $\langle b(h) : b \in B, h \in H\rangle$ is invariant, by reindexing b after any further action. The statement applies independently of any geometric conclusion about the quotient.

The induced action on A/N sends each image $A_i N/N$ to the image indexed by bi. Keep this indexed action as part of the data even if some subgroup images coincide. Every orbit-intersection condition involving the acting group and named types is consequently unchanged. The image subgroups are $\widetilde{A}_i = A_i N/N$, rather than A_i/N, which need not be defined. No flag transitivity, intersection property, firmness or residual connectedness is inferred for this quotient. Those are separate hypotheses when applying twisting.

6 Cyclic markings and exact presentations

6.1 The precise input

For a group generated by a finite named set S, write $A(R) = \langle r : r \in R\rangle$ for $R \subseteq S$, including $A(\varnothing) = 1$. We call this a parabolic marking if

$$A(R) \cap A(Q) = A(R \cap Q) \tag{14}$$

and the cosets of all $A(R)$ have the finite Helly property: every finite pairwise-intersecting collection has a common element. The names denote distinct nonidentity elements. The standard coset system has maximal parabolics $A(S \setminus \{s\})$.

Here are the classical inputs retained for an arbitrary Artin–Tits marking. Standard parabolic presentations embed, their intersections obey (14), and their cosets have this Helly property. These are the standard-parabolic results invoked in N005 Theorems 2.11–2.12: van der Lek's parabolic embedding and the intersection/product results cited there from Godelle–Paris. In particular, each singleton Artin

parabolic is infinite cyclic. For a Coxeter marking the corresponding facts are the classical Coxeter parabolic theorem, and singleton parabolics have order two. We do not reprove those external results or identify their proof costs with zero. Precisely, the source locators are Bourbaki, *Groupes et algèbres de Lie*, Chapters 4–6 (2006 edition), Chapter 4, §1, Theorem 2 and Exercise §1.1(b); H. van der Lek, *The homotopy type of complex hyperplane complements* (1983), Chapter II, Theorem 4.13; and E. Godelle–L. Paris, $K(\pi, 1)$ *and word problems for infinite type Artin–Tits groups, and applications to virtual braid groups*, Math. Z. 272 (2012), Proposition 4.5. We retain these interfaces as the primary source does, rather than claiming fresh proof verification of the entire cited works.

Lemma 1 identifies this Helly hypothesis with flag transitivity, once formula (14) identifies generated supports with standard intersections.

Lemma 26. *In a parabolic marking, for $H = A(R)$, $K = A(Q)$ and $L = A(P)$,*

$$HK \cap HL = H(K \cap L).$$

The complex of cosets of maximal parabolics, with simplices the families having common intersection, is a flag complex. Its coset geometry is residually connected; its panel of type s has $|\langle s\rangle|$ chambers.

Proof. If $z \in HK \cap HL$, the left cosets H, zK, zL meet pairwise: the first meets each of the others and the latter two contain z. Helly gives $h = zk = zl \in H$, with $k \in K, l \in L$; hence $k = l \in K \cap L$ and $z \in H(K \cap L)$. The reverse inclusion is immediate. The complex assertion is exactly Helly. Every standard residue has acting group $A(R)$ and maximal parabolics $A(R \setminus \{s\})$, $s \in R$. If $|R| \geq 2$ they generate $A(R)$: each named generator belongs to a child missing a different name. Its incidence graph is therefore connected (successive multiplications by parabolic elements give incidence paths). Flag transitivity identifies every residue with a translate of one of these. The Borel is 1 by (14), and the rank-one residue is $\langle s\rangle/1$, proving the panel assertion. □

6.2 One cyclic marking theorem

Let $S = Y \sqcup L$ with $L = \{x_0, \dots, x_{k-1}\}$ and $k \geq 2$. Suppose (A, S) is a parabolic marking and $\sigma \in \mathrm{Aut}(A)$ fixes every $y \in Y$ and sends x_j to x_{j+1}, with subscripts modulo k. Then $\sigma^k = \mathrm{id}$ since it fixes every generator. Set

$$G = A \rtimes C_k = \langle A, t \mid t^k = 1,\ t^{-1}at = \sigma(a)\ (a \in A)\rangle, \qquad T = Y \sqcup \{x_0, t\}.$$

This notation fixes the conjugation convention. The quotient onto C_k proves that t has exact order k. The set T generates G, since $x_j = t^{-j}x_0t^j$.

For $R \subseteq T$ define a subset $E(R)$ of S by

$$E(R) = \begin{cases} R, & t \notin R, \\ R \cap Y, & t \in R,\ x_0 \notin R, \\ (R \cap Y) \cup L, & t, x_0 \in R. \end{cases}$$

In the first line R is read inside $Y \sqcup \{x_0\} \subseteq S$.

Theorem 27. *For every $R \subseteq T$ the subgroup generated by R is*

$$G(R) = \begin{cases} A(E(R)), & t \notin R, \\ A(E(R)) \rtimes C_k, & t \in R. \end{cases} \tag{15}$$

The marking (G, T) has the intersection and coset Helly properties. Consequently its standard coset system is flag-transitive, residually connected and firm, and its coset complex is flag. The product identity of Lemma 26 holds for every three generated parabolics. Its panels have size k at t and $|\langle s\rangle|$ at every $s \in Y \sqcup \{x_0\}$.

Proof. If $t \notin R$, the assertion is the embedding of A in G. If $t \in R$ but $x_0 \notin R$, it fixes every named generator in $R \cap Y$, giving the second case. If both belong to R, conjugating x_0 produces exactly the whole orbit L, and this subgroup of A is invariant under t. This proves (15) without an order calculation or finiteness assumption on A.

If both subsets contain t, the two A-parts intersect by (14); the orbit L is present in their intersection precisely when x_0 belongs to both subsets. Thus their intersection is the subgroup for $R \cap Q$. If neither contains t, use (14) inside A. In the mixed case an element of the intersection has trivial C_k coordinate. The smaller support contains at most x_0 from L, so intersecting its support with the other support again gives exactly $E(R \cap Q)$. This proves full generated-subgroup intersection.

It remains to prove Helly, which does not follow from subgroup intersection alone. The maximal parabolics just computed are

$$P_t = A(Y \cup \{x_0\}), \quad P_{x_0} = A(Y) \rtimes C_k, \quad P_y = A(S \setminus \{y\}) \rtimes C_k \ (y \in Y). \tag{16}$$

The A-supports in the latter groups are invariant under C_k. Take pairwise-intersecting left cosets of these groups. Repeated types coincide. If a P_t coset occurs, translate its C_k coordinate to 1. All other cosets project onto C_k and have representatives $(a, 1)$. Their fibers at coordinate 1 are cosets of the indicated standard parabolics in A. Two cosets of the latter kind intersect if and only if their A fibers intersect, because they are Cartesian products with C_k; incidence with the P_t coset likewise takes place in this fiber. Thus the A fibers meet pairwise and base Helly gives a common representative. If no P_t coset occurs, the same argument applies at coordinate 1 without translation. This proves Helly for maximal parabolics. Replace a coset of any generated parabolic by its maximal-parabolic cosets, using the intersection property just proved, to get Helly for all of them. Finally apply Lemma 26. Every named generator is nontrivial and the panel sizes are at least two, proving firmness. □

This is the rank-one cyclic case of the orbit construction used in Piedade–Tranchida's papers on twisting. Its proof is included in full so that its geometric and presentation conclusions have a complete local proof. It does not assert that an arbitrary Shephard marking has these properties.

6.3 The involutory presentation and the doubled braid

Write $\mathsf{W}_m(u, v)$ for the alternating word of *length* m, starting with u. It is not the ordinary power $(uv)^m$. Let A be the Artin group on $Y \sqcup \{x, x'\}$, with Artin relations $\mathcal{R}_Y$ on Y, identical labels m_y on (x, y) and (x', y), and label m on (x, x'). All labels lie in $\{2, 3, ...\} \cup \{\infty\}$; an infinite label means no relation. Interchanging x, x' is an automorphism of the presented group, fixes Y, and has order two because the names are distinct by the retained Artin parabolic theorem.

Lemma 28. *If $t^2 = 1$, then for every $m \geq 1$,*

$$\mathsf{W}_m(x, txt) = (xt)^m t^{m \bmod 2}, \qquad \mathsf{W}_m(txt, x) = (tx)^m t^{m \bmod 2}.$$

Consequently the braid of length m on x, txt is equivalent to the braid of length $2m$ on x, t.

Proof. For $m = 2r$, the two words are $(xtxt)^r = (xt)^{2r}$ and $(txtx)^r = (tx)^{2r}$. For $m = 2r + 1$, append respectively x and txt to those words, obtaining $(xt)^{2r+1}t$ and $(tx)^{2r+1}t$. Cancellation of the common final t in the odd case proves the equivalence. The alternating words of length $2m$ are exactly $(xt)^m$ and $(tx)^m$. □

Theorem 29. *The semidirect product which interchanges x, x' has presentation*

$$\begin{aligned} \langle Y, x, t \mid \mathcal{R}_Y,\ t^2 = 1,\ [t, y] = 1 \ (y \in Y), \\ \mathsf{W}_{m_y}(x, y) = \mathsf{W}_{m_y}(y, x) \ (m_y < \infty), \\ \mathsf{W}_{2m}(x, t) = \mathsf{W}_{2m}(t, x) \ (m < \infty) \rangle. \end{aligned}$$

This is a Shephard presentation, and its named marking satisfies every conclusion of Theorem 27.

Proof. Map the displayed generators to the semidirect product. They satisfy every relator: the last by Lemma 28; the others follow from the defining action and the original Artin relations. They generate, since $x' = txt$. Conversely, in the displayed presentation set $x' = txt$. Conjugating a mixed braid by t, which fixes y, gives the braid on x', y. Lemma 28 supplies the braid on x, x'. The relations on Y are unchanged. Thus the original Artin presentation maps into this group, and conjugation by t implements its transposition of the two leaves. The semidirect-product universal property gives the inverse map. Both composites fix all named generators, so these maps are inverse. This establishes the exact group and marking, not merely a quotient action. The cyclic theorem with $k = 2$ now applies. An edge absent from the drawing has label two, so all commutations $[t, y] = 1$ remain part of the presentation. □

6.4 Cyclic amalgamation

Let A_0 be an Artin group on $Y \sqcup \{x\}$ and $C = A_0(Y)$. The parabolic embedding theorem identifies C with the Artin group on the induced graph on Y. Form the amalgam M of $k \geq 2$ copies of A_0 along this common C. Its presentation has Y and $x_0, \dots, x_{k-1}$, the relations $\mathcal{R}_Y$, and the same braid between x_j and each $y \in Y$. There are no relations between distinct x_j. This follows directly from the universal property of the amalgam; the embeddings of each factor are the classical normal-form theorem. The resulting presentation is itself Artin, with all distinct leaf-pair labels infinite.

Theorem 30. *Cyclic permutation of the copies defines an automorphism σ of M. With $t^{-1}x_j t = x_{j+1}$, its semidirect product has presentation*

$$M \rtimes C_k \cong \langle Y, x_0, t \mid \mathcal{R}_Y,\ t^k = 1,\ [t, y] = 1\ (y \in Y),\ \mathsf{W}_{m_y}(x_0, y) = \mathsf{W}_{m_y}(y, x_0)\ (m_y < \infty)\rangle.$$

There is no relation on the pair x_0, t. This Shephard marking has full generated-parabolic intersection and the product identity, and its standard coset system is flag-transitive, residually connected and firm with flag coset complex. Its panels have sizes k and infinity.

Proof. The permutation preserves the Artin relators and has order k on the distinct leaf names. The displayed generators map to the semidirect product and generate it. In the converse direction define $x_j = t^{-j}x_0 t^j$. Since t fixes Y, conjugation by t^j transports every mixed braid to the required braid on x_j, y. The relation $t^k = 1$ closes the cycle. The Artin presentation of M and then its semidirect product therefore map into the displayed group. Both composites fix Y, x_0, t, proving inverse maps. Apply Theorem 27 to the Artin marking of M. □

7 Marked permutation modules and changes of coordinates

Let $r \geq 2$, let V be an elementary abelian 2-group, regarded as an $\mathbb{F}_2[S_r]$-module, and let $z \neq 0$ be fixed by the stabilizer of 1 in S_r. Write $s_i = (i\ i+1)$ for $1 \leq i < r$. For each j put $z_j = \pi z$ for any π taking 1 to j. This is well-defined: two such permutations differ on the right by an element fixing 1. Let V_z be the span of these r vectors. In $V_z \rtimes S_r$ put

$$\rho_0 = (z, 1), \qquad \rho_i = (0, s_i) \quad (1 \leq i < r).$$

The multiplication convention is $(v, \pi)(w, \sigma) = (v + \pi w, \pi\sigma)$.

Theorem 31 (A marked module)**.** *The displayed marks form a string C-group of rank r, with generated group $V_z \rtimes S_r$. This assertion requires no linear independence of the orbit vectors $z_1, \dots, z_r$.*

Proof. All marks are nonidentity involutions. The marks ρ_i with $i \geq 1$ generate the indicated copy of S_r; their conjugates of ρ_0 generate V_z. If $i \geq 2$, then s_i fixes 1 and hence z, so ρ_0 commutes with ρ_i. The other string commutations are those of adjacent transpositions.

We prove every subgroup intersection explicitly. For a set of mark indices $D \subseteq \{0, \dots, r-1\}$ let $E = D \setminus \{0\}$ and $P_E = \langle s_i : i \in E \rangle$. This is the product of the symmetric groups on the connected components of the path with retained edges E: path transpositions generate each component and move no point outside it. Its orbit on 1 is an initial segment $[k(E)]$, where $k(E)$ is one plus the length of the initial run $1, 2, \dots$ contained in E. Set

$$W_D = \begin{cases} \langle z_1, \dots, z_{k(E)} \rangle_{\mathbb{F}_2}, & 0 \in D, \\ 0, & 0 \notin D. \end{cases}$$

The group generated by D is exactly $W_D \rtimes P_E$. Indeed, all its vector generators are conjugates of z under P_E, and those conjugates are precisely the vectors in the displayed span. The other path components fix z and do not introduce any further vector.

For two edge sets E, F, a permutation preserving each block of both component partitions preserves each of their pairwise intersections. Those intersections are exactly the components of the path retaining $E \cap F$: two vertices are in the same component precisely when every edge of their unique interval belongs to both sets. Thus $P_E \cap P_F = P_{E \cap F}$. Also $k(E \cap F) = \min\{k(E), k(F)\}$. The vector spaces spanned by initial segments form a nested chain, even when some orbit vectors are dependent. Consequently $W_D \cap W_{D'} = W_{D \cap D'}$. Intersecting inside the Cartesian set $V_z \times S_r$ proves the full C-group property. □

The elementary coordinate change below applies to any group action on an abelian group, not only to permutation modules.

Lemma 32 (Cocycle change of coordinates)**.** *Let Q act on an abelian group V, written additively. Suppose $c : Q \to V$ satisfies*

$$c(gh) = c(g) + gc(h).$$

Then $\Phi_c(v, g) = (v + c(g), g)$ is an automorphism of $V \rtimes Q$, with inverse Φ_{-c}. It fixes V pointwise and carries any marked subgroup intersection identity to the corresponding identity on the image marks. In characteristic two it is an involution.

Proof. The cocycle identity with $g = h = 1$ gives $c(1) = 0$. For two elements,

$$\Phi_c((v, g)(w, h)) = (v + gw + c(g) + gc(h), gh) = \Phi_c(v, g)\Phi_c(w, h).$$

Addition of $-c(g)$ in the same coordinate is an inverse. The last assertions follow from $c(1) = 0$, from bijectivity, and from $c(g) + c(g) = 0$ in characteristic two. In particular an automorphism commutes with taking generated subgroups and their intersections. □

Proposition 33 (Three pairs of signed markings)**.** *Take $V = \mathbb{F}_2^r$ with its natural permutation action. Write $e_1, \dots, e_r$ for its basis and $\lambda = e_1 + \cdots + e_r$. For any of the three seeds*

$$z = \lambda, \qquad z = e_1, \qquad z = \lambda + e_1,$$

both of the following lists are string C-group markings:

$$(z, 1), (0, s_1), \dots, (0, s_{r-1}), \qquad (z, 1), (\lambda + e_1 + e_2, s_1), \dots, (\lambda + e_{r-1} + e_r, s_{r-1}). \tag{*}$$

The common abstract groups for these three pairs are respectively $C_2 \times S_r$, $C_2 \wr S_r$, and

$$\begin{cases} C_2 \wr S_r, & r \text{ even}, \\ V^{\text{even}} \rtimes S_r, & r \text{ odd}, \end{cases} \qquad V^{\text{even}} = \{v : \sum_j v_j = 0\}.$$

In the last line $r \geq 3$ when odd. The statement includes the odd-rank second list for the third seed.

Proof. Each seed is nonzero and fixed by the stabilizer of 1, so the first list is covered by Theorem 31. To identify its vector span, only the third seed needs explanation. Its orbit vectors are $\lambda+e_j$. A relation with coefficients b_j says $b = (\sum_j b_j)\lambda$. For even r this forces $b = 0$; for odd r the only nonzero relation is $b = \lambda$. In the odd case every orbit vector has even coordinate sum, so their $(r-1)$-dimensional span is exactly V^{even}.

Let $a \in V$ have alternating coordinates, so that $a_i + a_{i+1} = 1$. Let $\epsilon : S_r \to \mathbb{F}_2$ be the sign homomorphism. Define

$$c(\pi) = \epsilon(\pi)\lambda + a + \pi a.$$

The first summand is a cocycle because λ is fixed and ϵ is additive. The other two are a coboundary: $a + \pi\sigma a = (a + \pi a) + \pi(a + \sigma a)$. For s_i the result is $c(s_i) = \lambda + e_i + e_{i+1}$. Lemma 32, in the full ambient $V \rtimes S_r$, therefore carries the first list in (*) to the second, and fixes its first mark. It need not preserve the first list's named subgroup as a subset; it is an isomorphism onto the subgroup generated by the second list. This distinction matters for odd r, where the image copy of S_r may have odd vector coordinates. It does not affect the intersection property or the asserted abstract group. □

Source correspondence and attribution. The ordinary lists in Proposition 33, in the order of its three seeds, are graph (1) of Table 9 and graphs (13), (14) of the cited Fernandes–Piedade catalogue. The changed-coordinate lists are graph (2) of Table 9 and graphs (15), (16), respectively. In the bound arXiv:2407.16003v2 edition the latter four diagrams are in Table 6, physical page 25; the target paper variously calls that table 5 or 6. The catalogue's Proposition 4.7 and Table 6 concern $r = n/2 \geq 7$; the algebraic marking theorem proved here holds for every $r \geq 2$. Here a vector e_j flips the two points in block j, and s_i interchanges adjacent blocks without flips. Thus the vector $\lambda+e_i+e_{i+1}$ means precisely the vertical i-edges on every block except i and $i+1$, as shown in those diagrams.

The known family statements are Piedade–Tranchida [2, Proposition 5.1]. The odd-rank graph (16) is expressly left unverified by their method in the paragraph immediately before that proposition. The formula above proves that case as well. The source bridge and the full proof were distinctly reviewed in the precursor edition; the present migration retains both. This is not a claim that no previous work settles it. Cocycle changes of splitting are standard group cohomology; the contribution here is their explicit use to identify the paired markings, together with the nested-orbit-span proof which tolerates the odd-rank relation. No source-specific theorem is presumed in either proof.

8 Attribution and retained interfaces

The target constructions are due to Claudio Alexandre Piedade and Philippe Tranchida [1, 2, 3]. The tree support and local-coset arguments, indexed qualifications, cyclic presentation maps and module-coordinate proofs here reorganize the stated results and supply explicit corrections; no historical priority is asserted. Normal forms and the Bass–Serre tree construction are retained from classical combinatorial group theory, including Lyndon–Schupp, *Combinatorial Group Theory*, Chapters I and IV, and Serre, *Trees*. The Artin and Coxeter parabolic interfaces are stated with exact locators in the cyclic section. Their complete external proofs are not reproduced and their costs are not set to zero.

The existing `OrbitTwisting`, `PrefixSpans` and `CocycleShear` Lean files certify only their declared algebraic interfaces. They do not formalize the new tree, HNN, cyclic Helly, reflection or affine arguments. This edition adds no Lean declaration. Mathematical and formal verification are separate boundaries.

References

[1] C. A. Piedade and P. Tranchida, *From Group Operations to Geometric Structures: Amalgamations, HNN-Extensions, and Twisting in Coset Geometries*, arXiv:2505.24662v1 (2025).

[2] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, revised edition (2026).

[3] C. A. Piedade and P. Tranchida, *Coset geometries acting on their elements: The j-diagonals twisting*, arXiv:2608.23223v1 (2026).

[4] M. E. Fernandes and C. A. Piedade, *Regular polytopes of rank $n/2$ for transitive groups of degree n*, arXiv:2407.16003v2 (2025); J. Algebra 689 (2026), 256–296. The existing signed-family adapters use Tables 6 and 9 of the bound arXiv edition.